

Quick Overview to The JavaScript

JavaScript array provide variety of method to process the array: -

Sr No	Method	Purpose
Basic Methods		
1	toString()	Converts array into the String
2	at(index)	Returns element of given index
3	Join('*')	It is similar to the toString(), but in addition you can specify the separator.
4	pop()	It removes the last element from array, and at the same time it returns the value of removed element.
5	push(element)	Push method adds new element in the array at end and returns the length of array including pushed element.
6	shift()	Shift method removes the first element from array and shifts all remaining array elements to the lower index.
7	unshift("element")	unshift method add new element to the beginning of an array, and increate index of remaining element by one position
8	delete array[i]	delete make empty hole to the index not remove the index.
9	concat(...args)	Concat method merger two or more arrays without changing them, and produce/return new array



10	flat()	Flat method create new array by concat sub-array (inner-array) elements.
11	splice(spliceAt,removeNo, newItem)	Splice method adds/remove new elements in array to the desire index. It returns removed elements array if any.
12	Slice(startIndex , endIndex+1)	slice method returns part of an array from given start index to the given end index
Search Methods		
1	indexOf(element)	indexOf method returns index number of given array element at first occurrence.
2	lastIndexOf(element)	lastIndexOf method returns index number of given array elements at last occurrence.
3	includes(element)	It return true if given element is present in array
4	find (testFunction)	find method returns the first element which can satisfy the givent test function.
5	findIndex(textFunction)	findIndex method returns the index of first element which can satisfy the given test function
6	findLast(textFunction)	findlast method returns the last element which can satisfy the givent test function.
7	findLastIndex(textFunction)	findLastIndex method returns the index of last element which can satisfy the given test function
Sorting Methods		
1	Sort()	sort method sorts the array elements alphabetically



2	<code>reverse()</code>	reverse method reverses the elements of array, it is use to sort array in descending.
3	<code>toSorted()</code>	It returns new array by sorting existing array
4	<code>toReversed()</code>	It returns new array by reversing existing array
5	<code>sort(function)</code>	To sort according to numeric property
Iteration methods		
1	<code>forEach(function)</code>	forEach method calls call back function once for each element of array,
2	<code>map(function)</code>	map method returns new array by performing function on each element of an existing array
3	<code>filter(function)</code>	Filter function returns the new array with the elements, which pass the test function

Example: -

```
let days=['SUN','MUN','TUE','WED','THU','FRI','SAT'];

console.log(days);

let paragraph=document.getElementById("para1");
paragraph.innerHTML=days

console.log(`days array have ${days.length} elements`)

console.log("0th index :- "+days[0])
console.log("1st index :- "+days[1])
console.log("2nd index :- "+days[2])
```



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Complete Java Classes

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```
console.log("3rd index :- "+days[3])
console.log("4th index :- "+days[4])
console.log("5th index :- "+days[5])
console.log("6th index :- "+days[6])

let stringDays=days.toString();
console.log(stringDays)

let value= days.at(3);
console.log("3rd index value :- "+value)

let seprateDays= days.join("/");
console.log(seprateDays);

console.log("Old Array:- "+ days);
let removedValue=days.pop()
console.log("Array after Pop:- "+days)
console.log("Removed element:- "+removedValue)
let oldLength=days.length
let newLength =days.push("Saturday");
console.log("Array after Push:- "+days)
console.log(`Old length is ${oldLength} and new length ${newLength}`)

days.shift()
console.log("Array after shift:- "+days);
days.unshift("Sunday");
console.log("Array afer unshift:- "+days)

delete days[3]
console.log("Array afer delete:- "+days)

let years=[2010,2020,2030,2040,2050];

let daysAndYears= days.concat(years,['sagar','shubham','sham']);
console.log("New merge Array:- "+daysAndYears);
```



```
let numbers=[10,40,30,20,10,70,50,90,30,80,60];  
let firstOccur=numbers.indexOf(30);  
let lastOccur=numbers.lastIndexOf(30);  
console.log('First occurrence index:- '+firstOccur);  
console.log('Last occurrence index:- '+lastOccur);
```

❖ Constant Object and Array: -

The keyword const is little misleading, it does not define the constant value, it defines the constant reference.

Because of this you cannot: -

1. Reassign a constant value.
2. Reassign a constant array.
3. Reassign a constant object.

But you can: -

1. Change the elements of constant array.
2. Change the properties of constant objects.